



Neural Signal Processing & Neuro-rehabilitation

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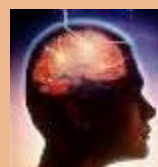
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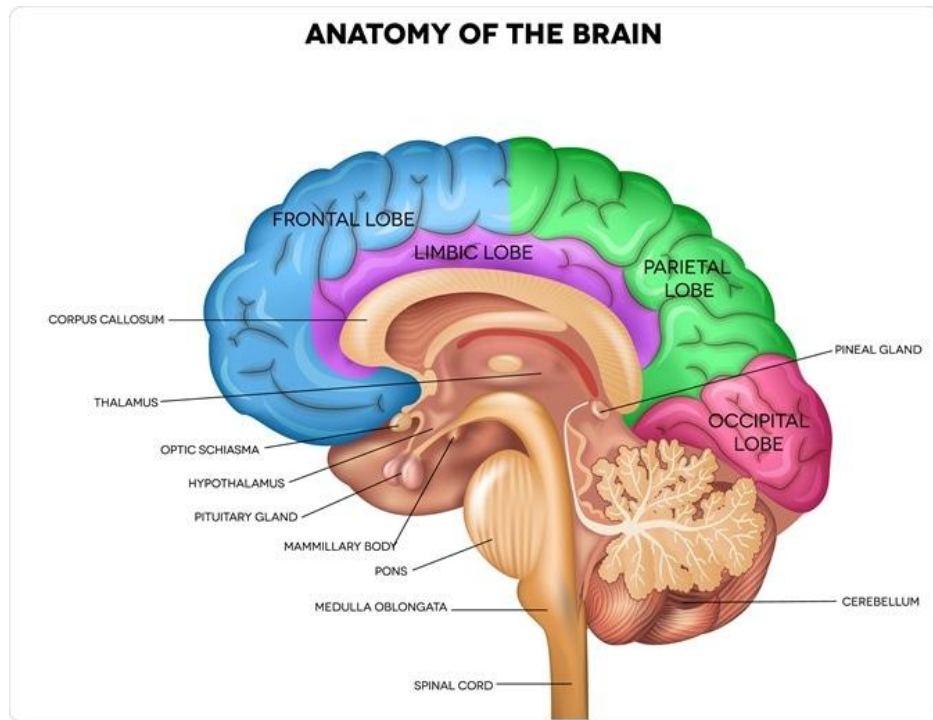
ulster.ac.uk



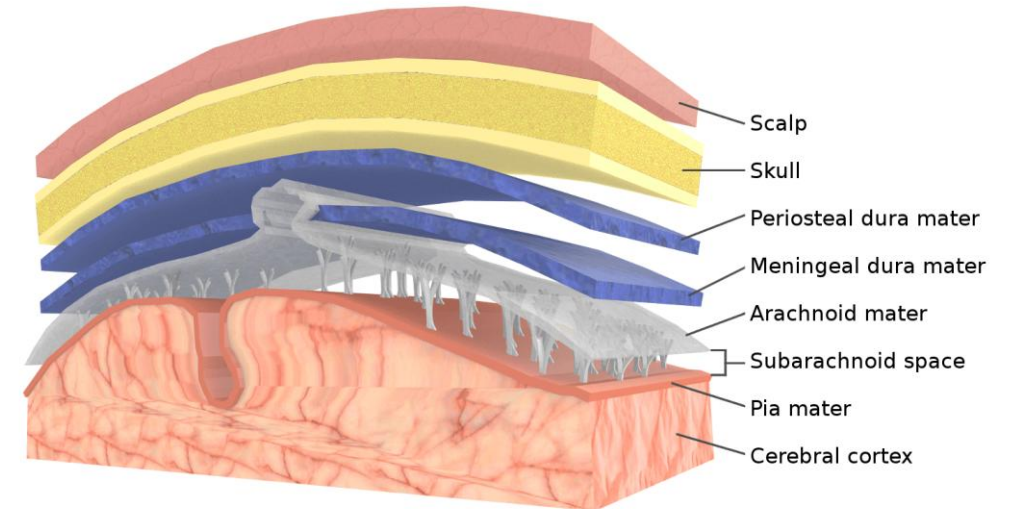
Part-1: Neural Signal Processing

Courtesy Dr Saugat Bhattacharyya

Measuring Brain Activity



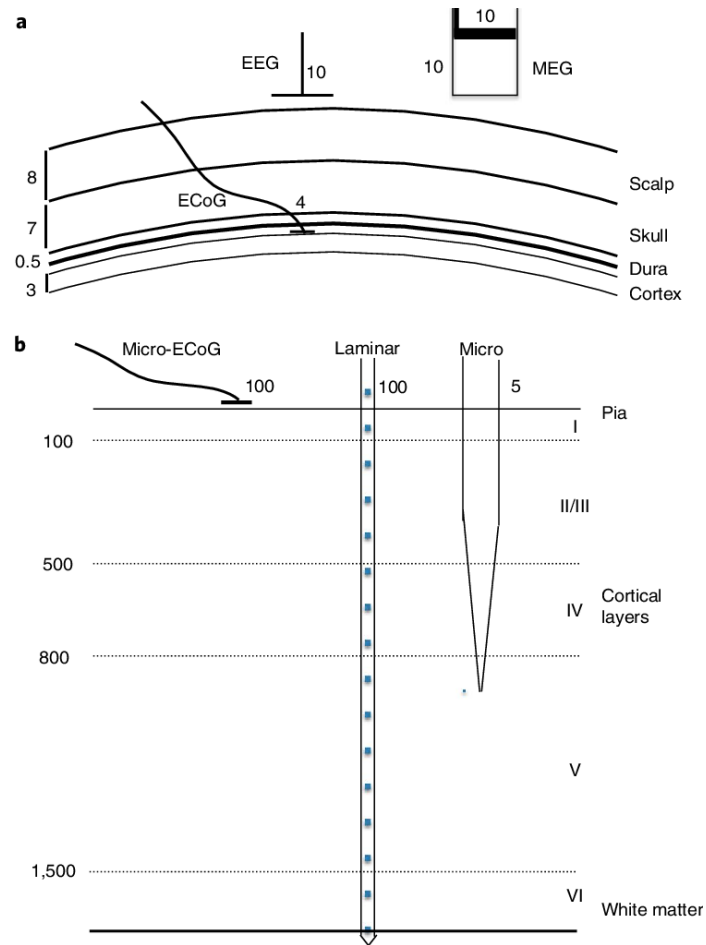
<https://www.news-medical.net/health/The-Anatomy-of-the-Human-Brain.aspx>



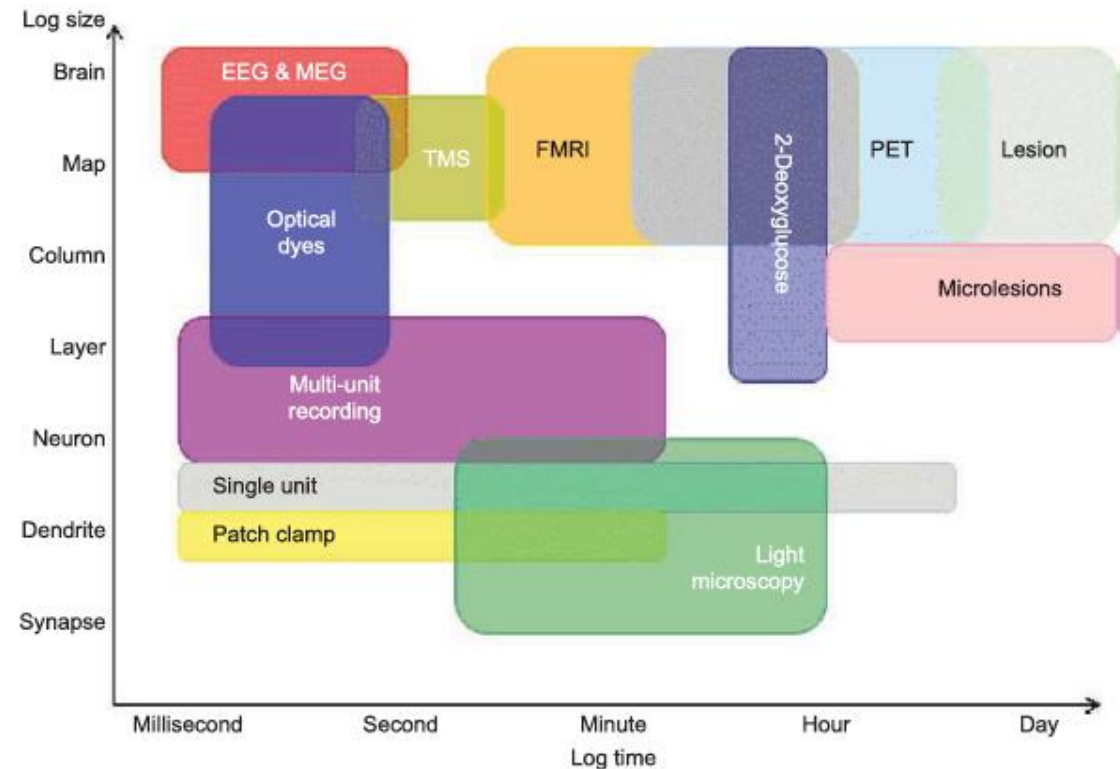
<https://www.thoughtco.com/brain-anatomy-meninges-4018883>

ETB 2016

Measuring Brain Activity

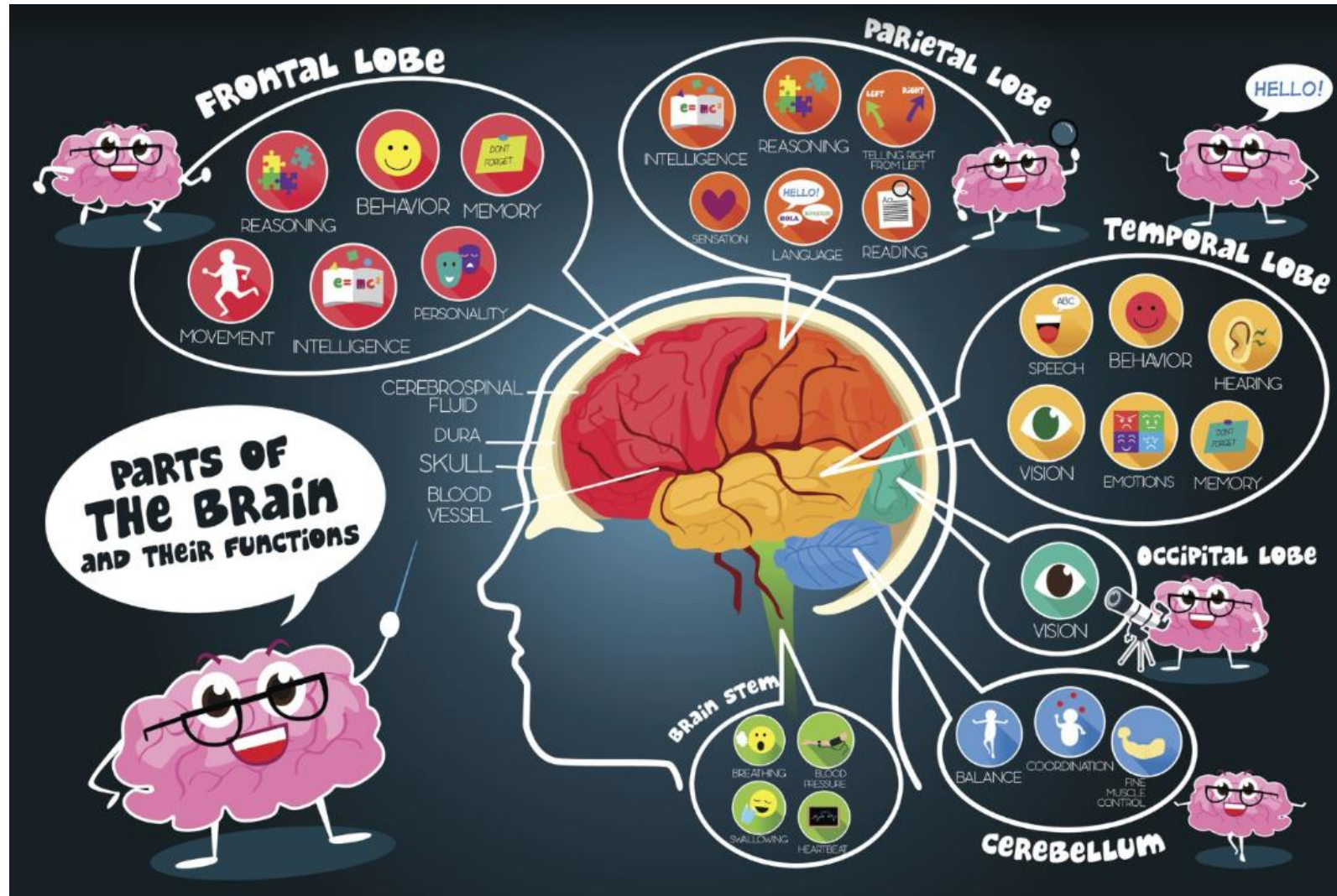


Pesaran (2018) Nature Neuroscience



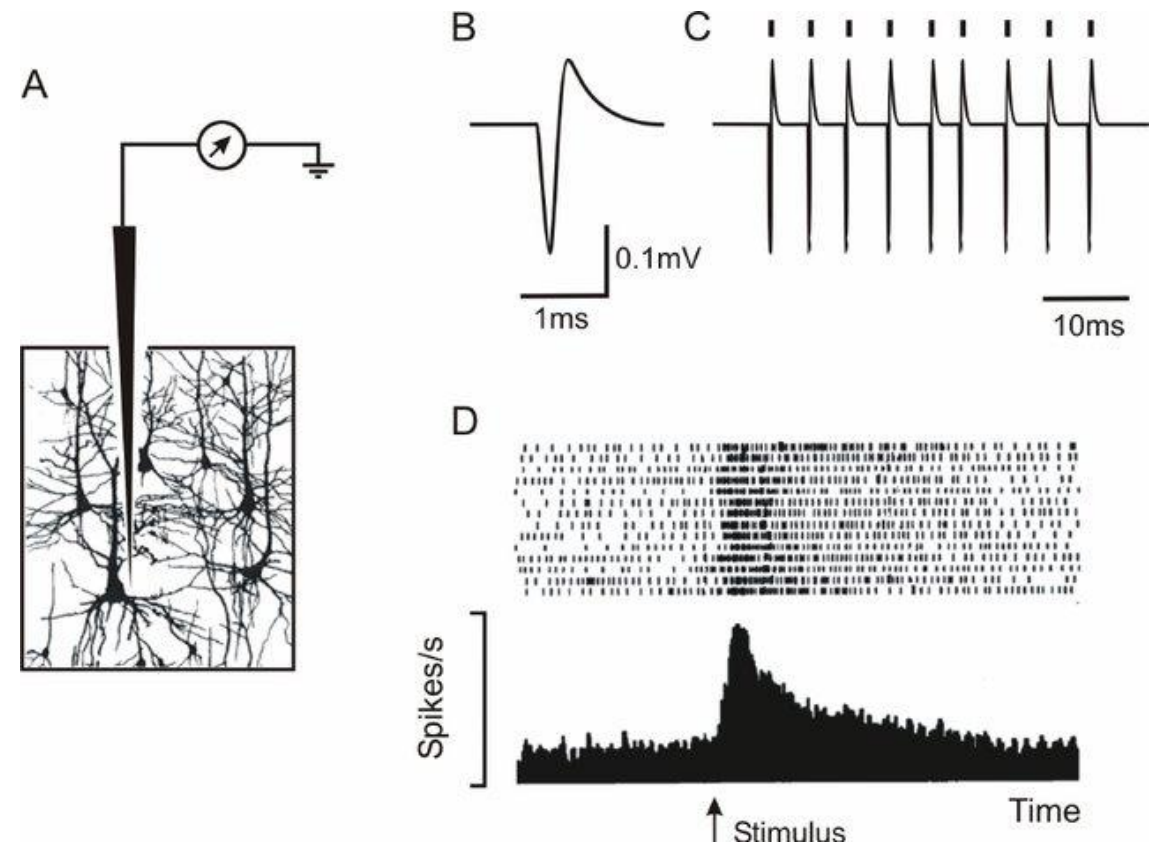
Ramsey et al. (2010) Cognition, Brain & Consciousness

Why Measure Brain Activity?



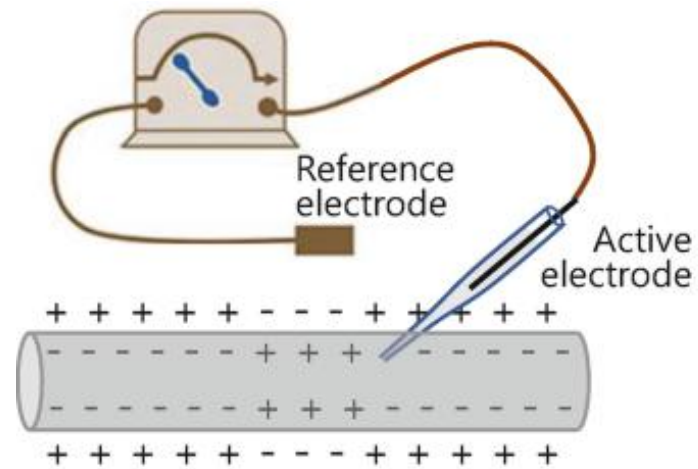
Single-unit Recording

- It is the use of an electrode to record the electrophysiological activity (action potentials) from a single neuron.
- An electrode introduced into the brain will detect electrical activity that is generated by the neurons adjacent to the electrode tip. If the electrode is a microelectrode, with a tip size of 3 to 10 micrometres, the electrode will often isolate the activity of a single neuron.
- The activity consists of the voltages generated in the extracellular matrix by the current fields outside the cell when it generates an action potential. Recording in this way is generally called "single-unit" recording.

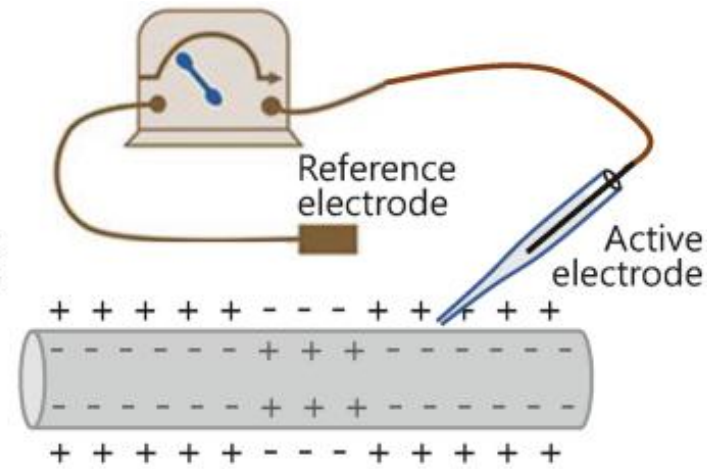


Ludvig (2011) Computational Neuroscience for Advancing Artificial Intelligence: Models, Methods and Applications

Intracellular & Extracellular Recording



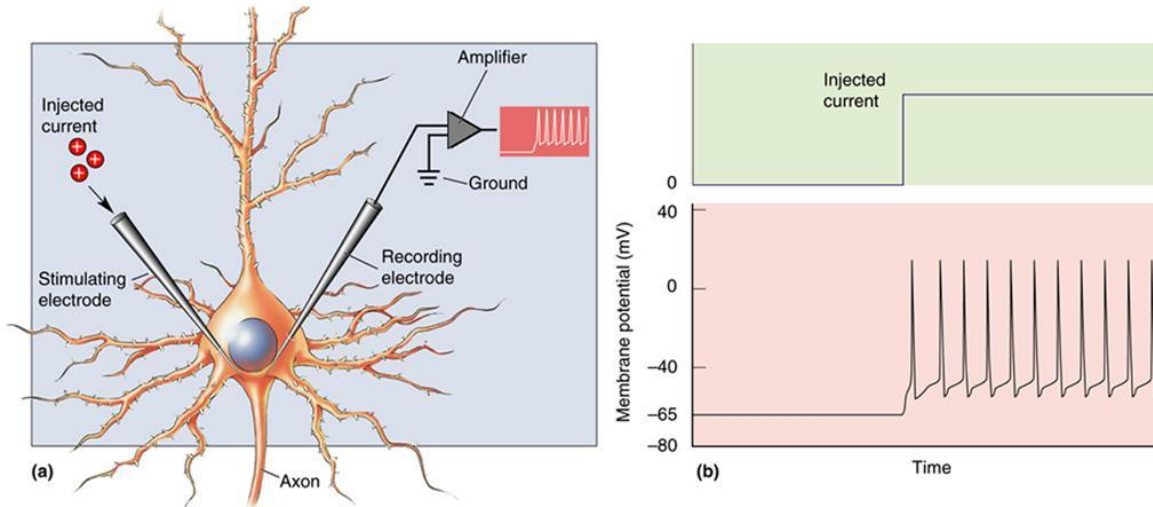
Intracellular recording



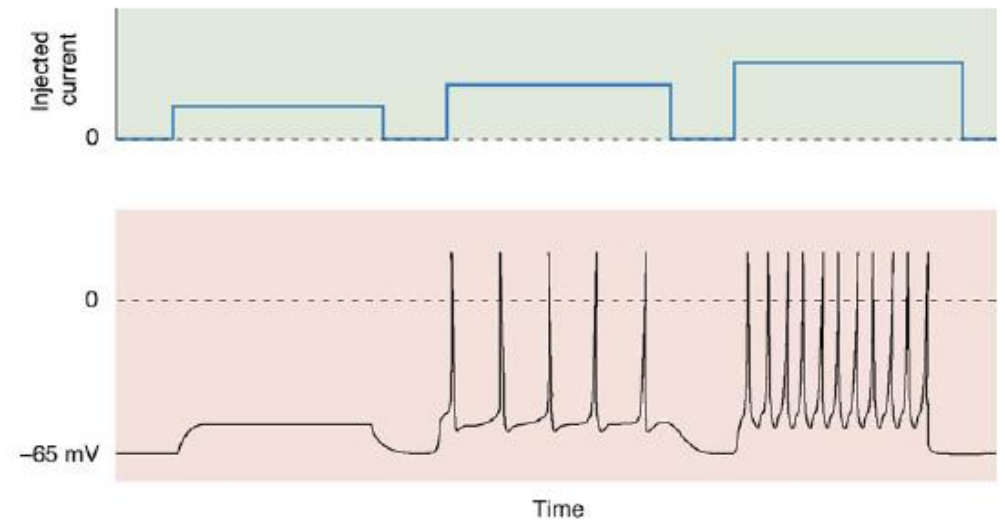
Extracellular recording

Measuring Action Potentials

Artificially inject current into a neuron with microelectrode



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- **Firing frequency** reflects the magnitude of the depolarizing current

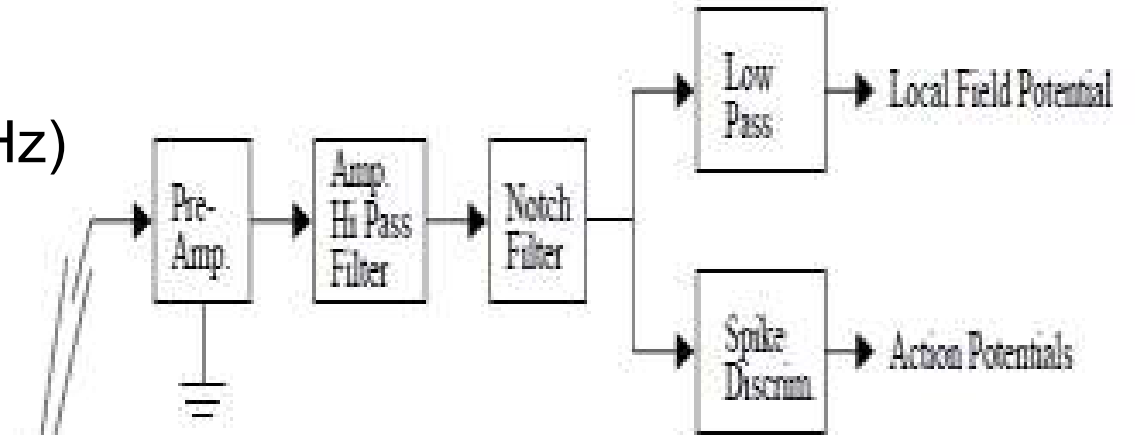
If injected current does not depolarize the membrane to threshold, no action potentials will be generated.

If injected current depolarizes the membrane beyond threshold, action potentials will be generated.

The action potential firing rate increases as the depolarizing current increases.

Local Field Potential

- Neural recordings from intracortical extracellular microelectrodes.
- Two components super-imposed:
 - Action potential/Spikes (300Hz – 5kHz)
 - Slow Varying Field Potential (1Hz – 250Hz)
- Represents the summation of excitatory and inhibitory dendritic signals and other types of slow activity (voltage dependant membrane oscillations or spike after-potentials)



Frequency Spectrum

- **5-50 μV** , mostly below 30 μV
- Sharp spike-waves, **light sleep** stages
- **5-120 μV** , mostly below 50 μV
- **Awake**, eyes closed, mental inactivity, physical relaxation
- **20-200 μV**
- Strictly rhythmic or highly irregular
- Awake & drowsiness, **light sleep** stages
 - LTP and phase-encoding
- **5-250 μV**
- Abnormality in waking adults, accompaniment of **deep sleep**

+ Gamma waves?
31-100 Hz, 10 μV
 'binding of consciousness',
 unity of perception

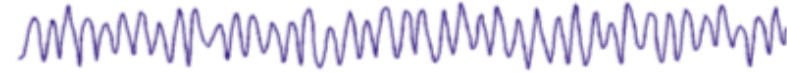
Beta (β) 13-30 Hz

Frontally and parietally



Alpha (α) 8-13 Hz

Occipitally



Theta (θ) 4-8 Hz

Children, sleeping adults



Delta (δ) 0.5-4 Hz

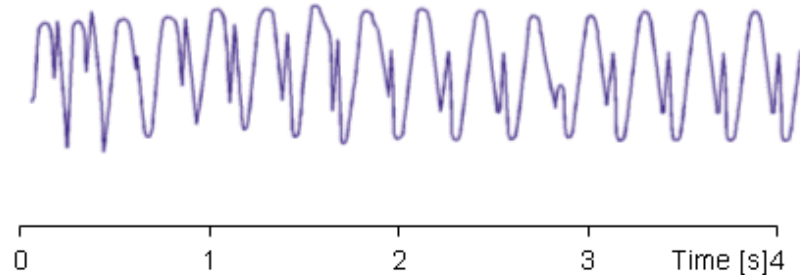
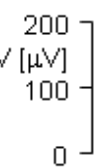
Infants, sleeping adults



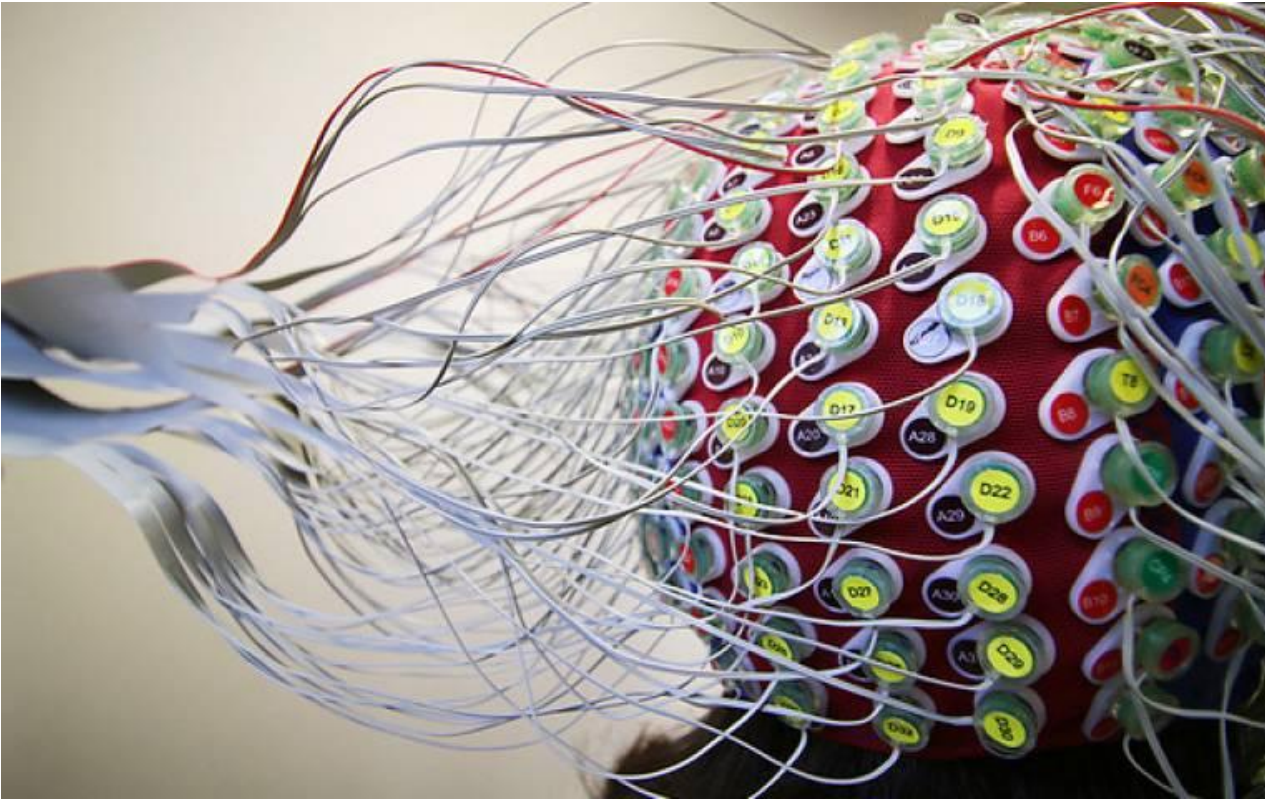
Spikes 3 Hz

Epilepsy - petit mal

V [μV]

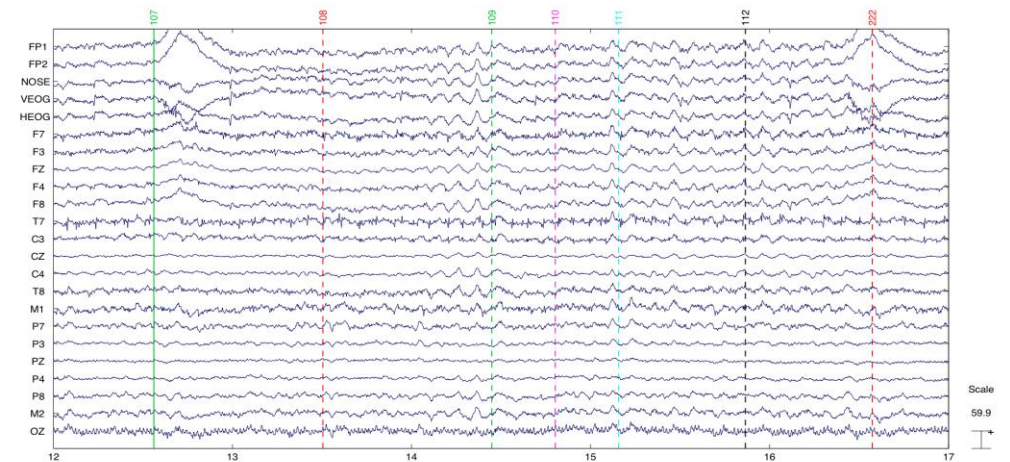


Electroencephalography



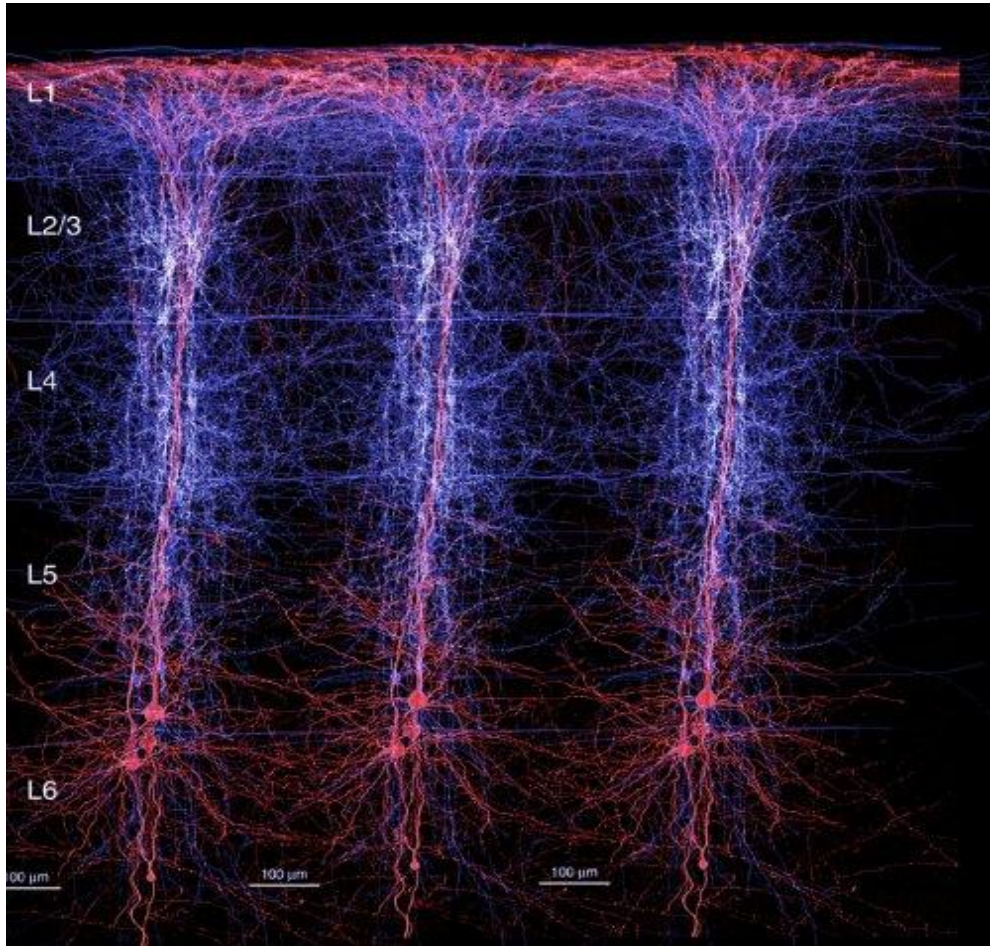
<https://www.ucl.ac.uk/brain-sciences/news/2023/jan/brain-wave-recordings-could-reveal-cause-catatonia>

- Electroencephalogram (EEG) electrodes
- Scalp recording of electrical activity of cortex => waveform signals
- Microvolts (μV) – small!



Electroencephalography

Neural Basis

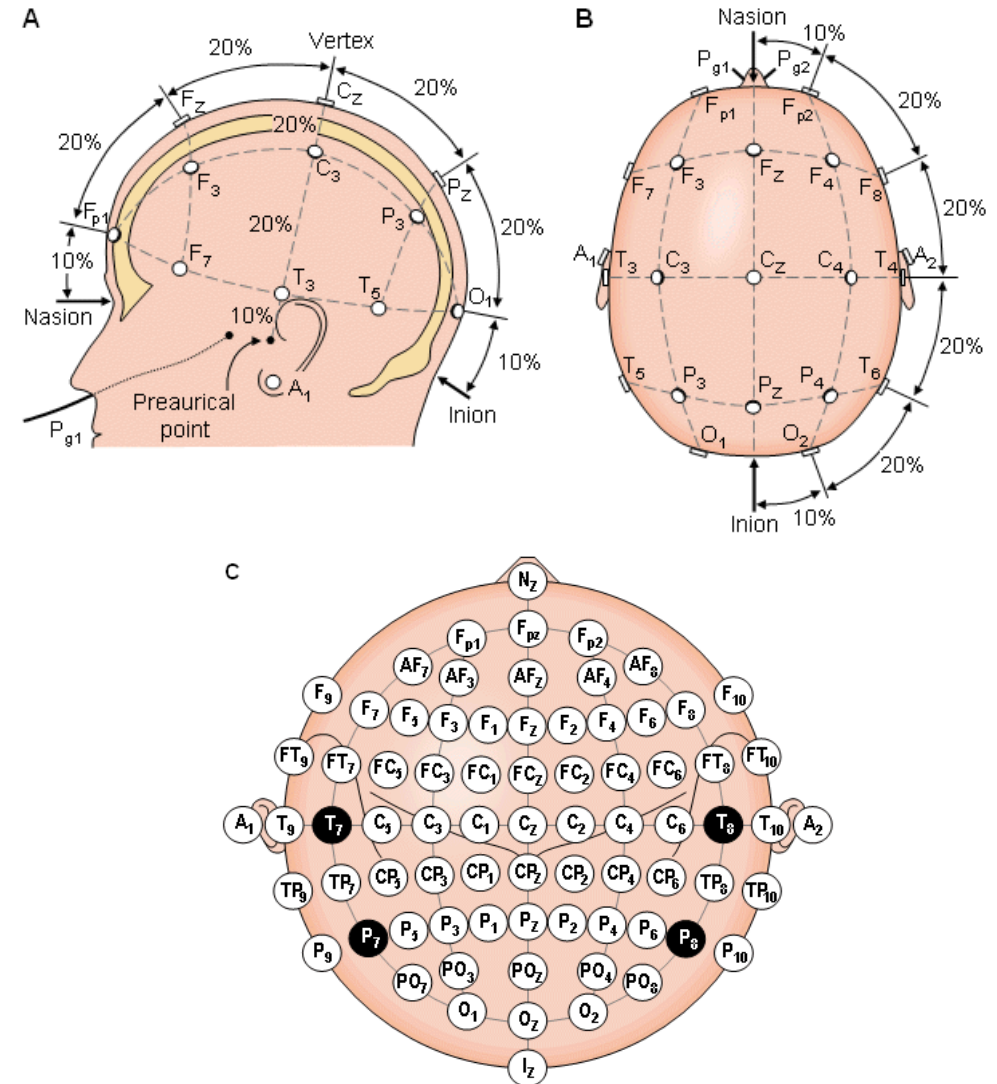


- Pyramidal neurons are spatially aligned and perpendicular to the cortical surface.
- Thus, EEG represents mainly the postsynaptic potentials of pyramidal neurons close to the recording electrode.
- The electrical activity from deeper generators gets dispersed and attenuated by volume conduction effects.

Electroencephalography

Surface Recordings

- **International 10/20 or 10/10 system** for placing electrodes:
 - **A**: anterior, **C**: central,
 - **P**: parietal, **F**: frontal,
 - **O**: occipital
- Low impedance 5-10k Ω
- Record **montages**:
 - **Bipolar** (electrodes connected to each other)
 - **Referential** (electrodes connected to one reference)



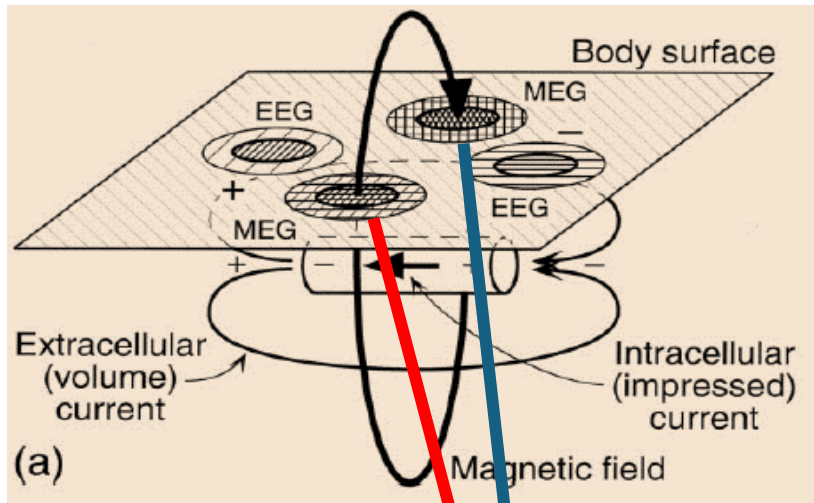
Magnetoencephalography



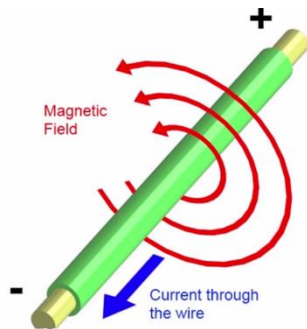
<http://www.admin.ox.ac.uk/estates/capitalprojects/previouscapitalprojects/megscanner/>

- Direct external recordings of **magnetic fields** created by electrical currents in cortex
- Measured in **fT** to **pT**
- Role of MEG in neuroimaging:
 - **Neural correlates** of cognitive/perceptual processes
 - **Localise** affected regions before surgery(?), determine regional and network functionality

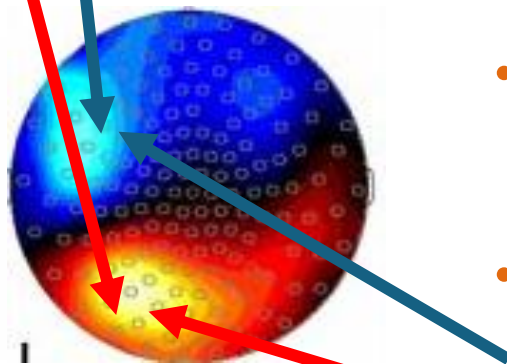
Magnetoencephalography



Tiege & Zlobinski, 2006



<http://www.youtube.com/watch?v=CPj4jJACels>

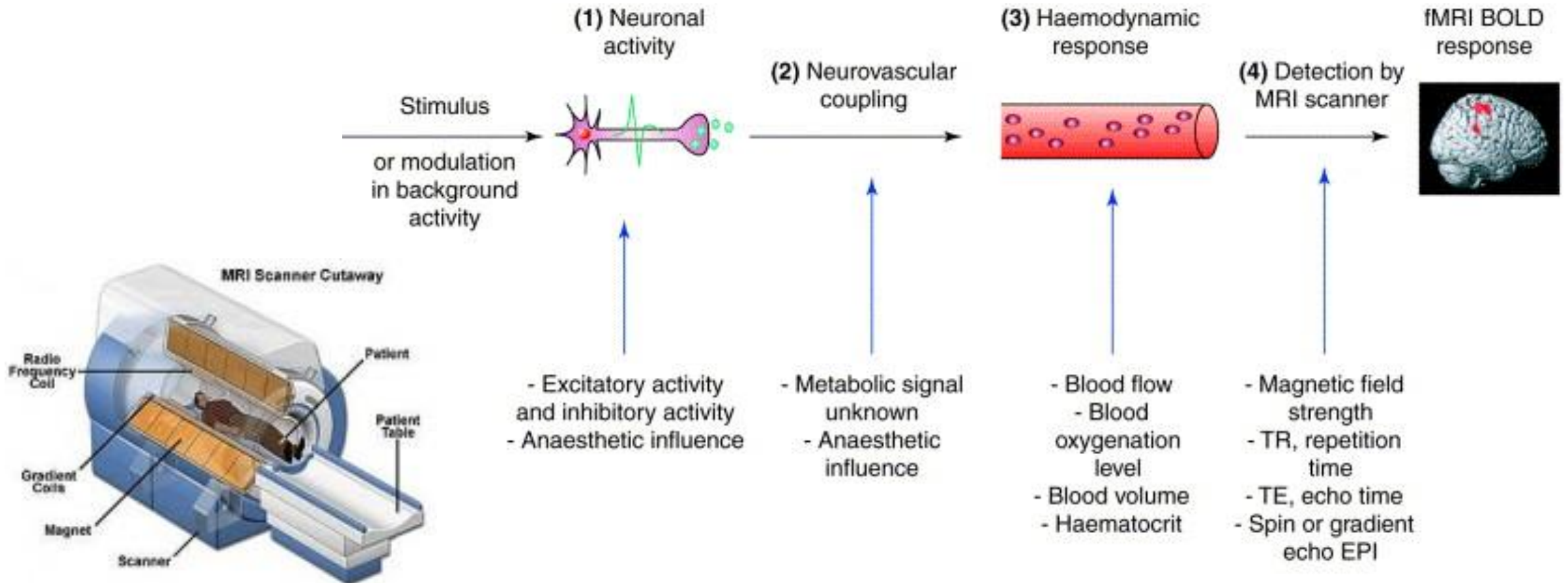


Ochi et al. 2011

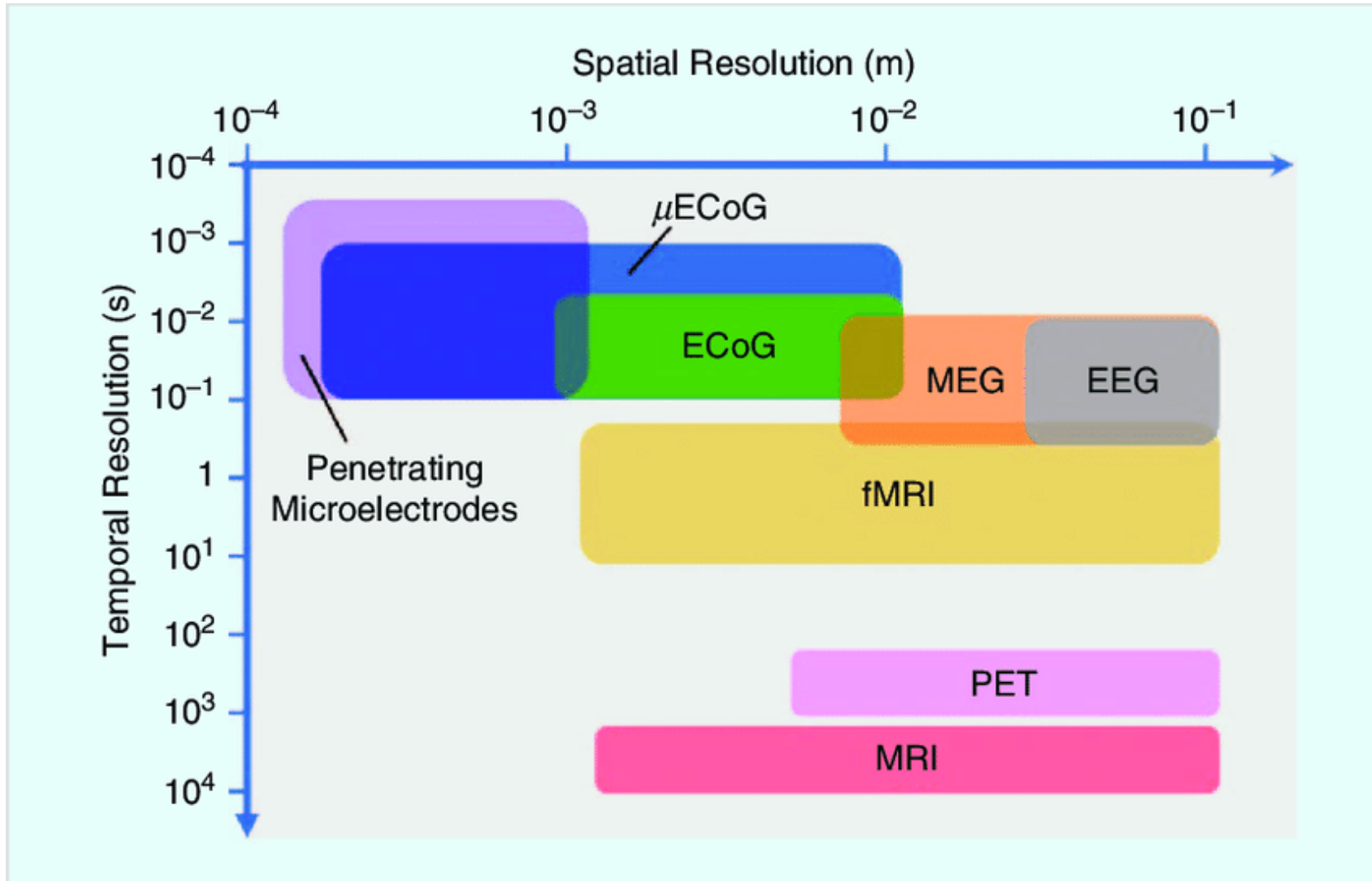
- **Large pyramidal neurons** in layer V of cortex, arranged in parallel, similarly-oriented, perpendicular to surface, fire synchronously
- Dipolar current flow generates a **magnetic field**.
- **10,000 to 50,000** active neurons required for detectable signal
- **Scalp topography:**
 - **Influx maxima** 'source'
 - **Efflux maxima** 'sink'

Functional MRI

- Measuring BOLD

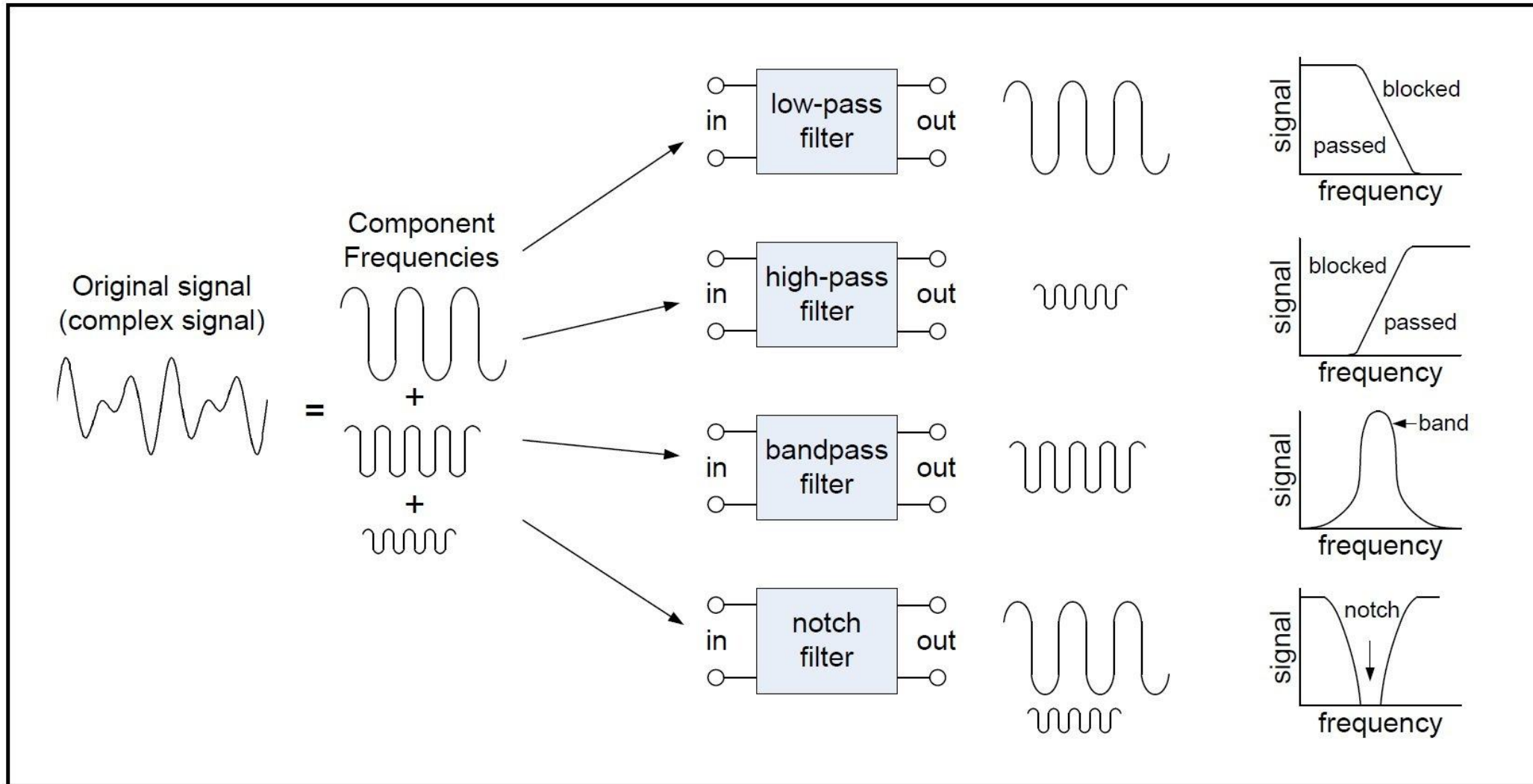


Spatial & Temporal Resolution



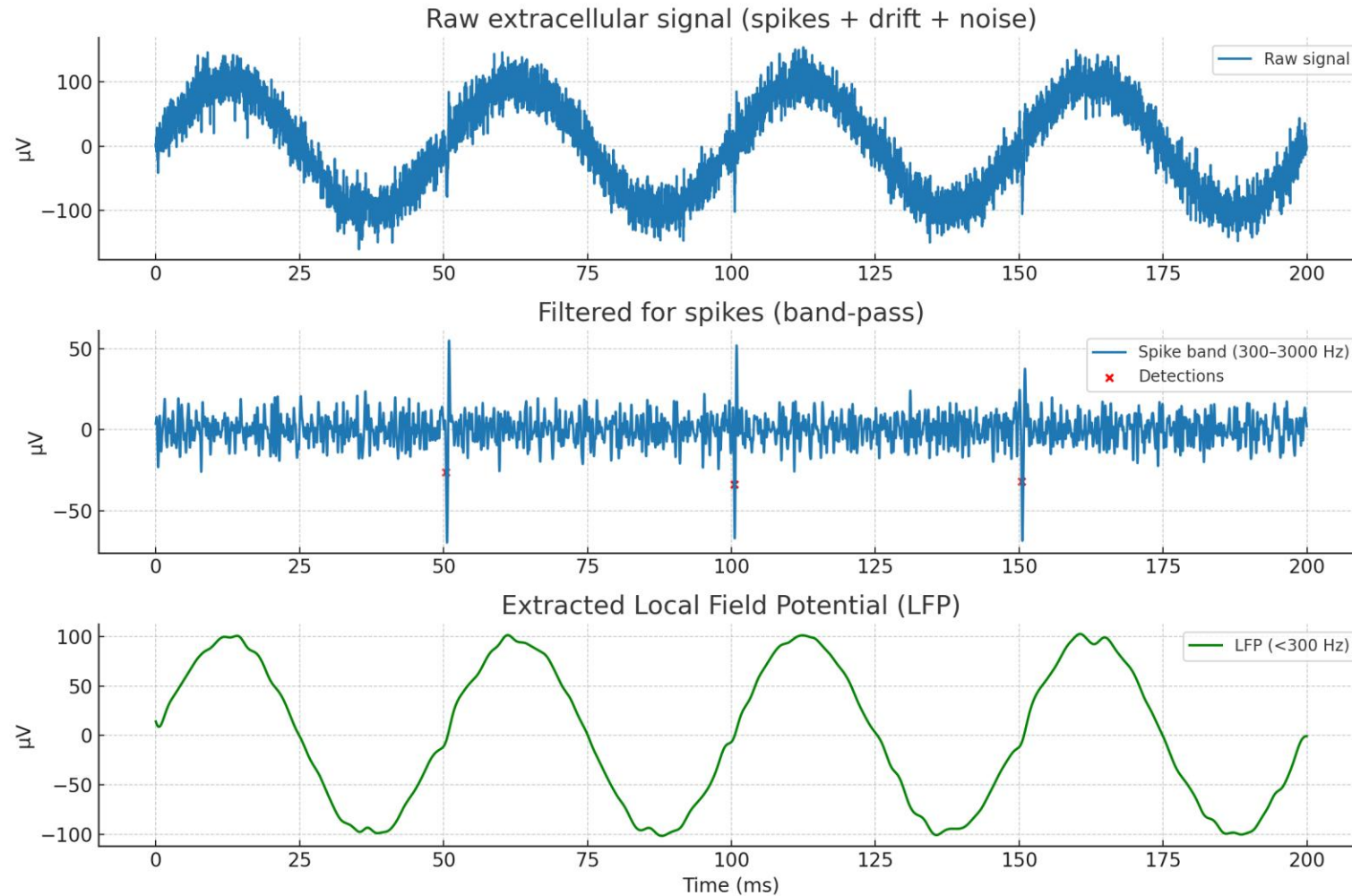
Common Pre-processing Approach

Temporal Filters



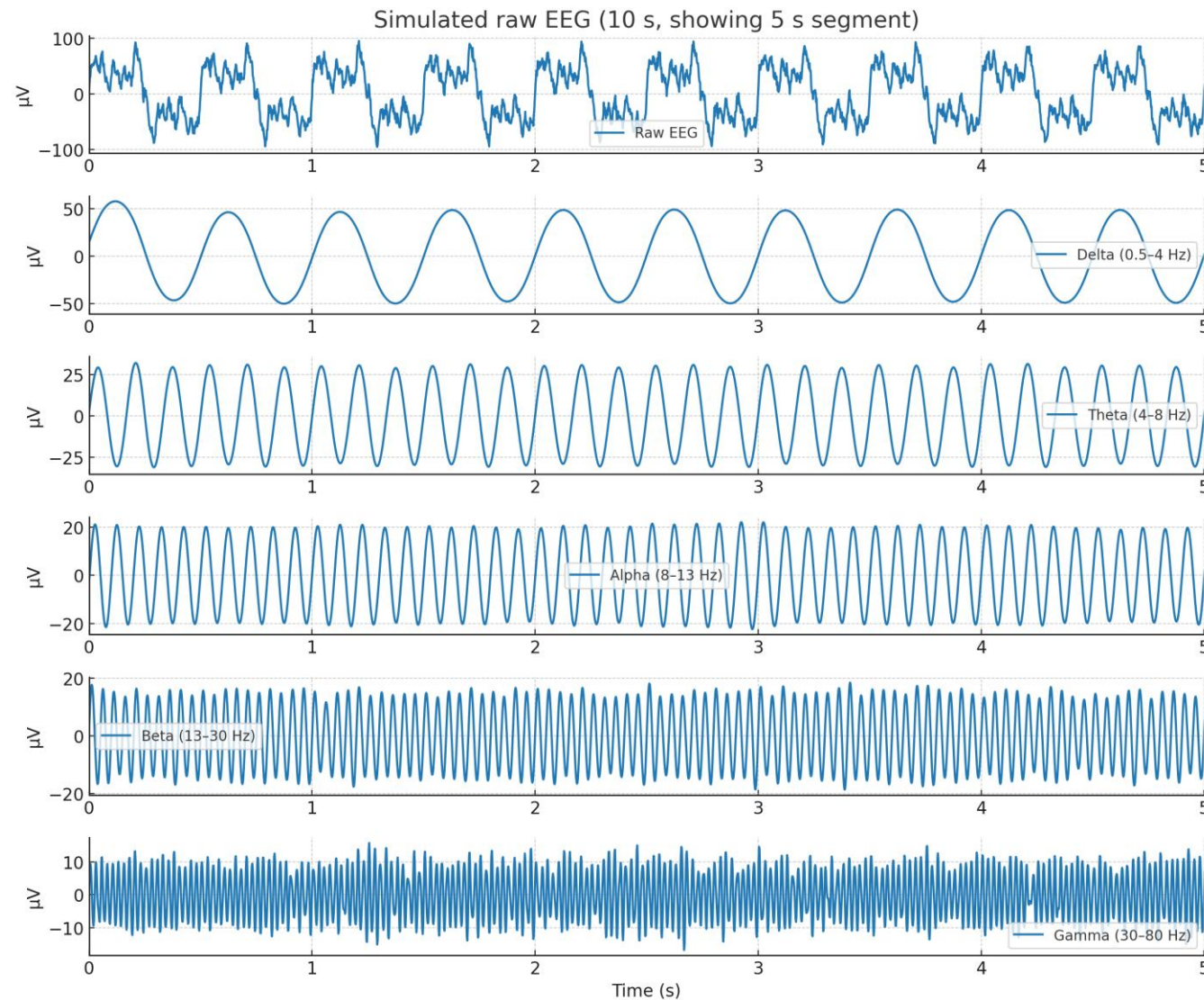
Temporal Filters

- Spike Detection and LFPs



Temporal Filters

- EEG



Spatial Filters

- **Bipolar:** the voltage difference between two electrode pairs

- **Laplacian**

$$V_i^{Lap} = V_i^{ER} - \sum_j g_{ij} V_j^{ER} \quad \text{where} \quad g_{ij} = \left(d_{ij} \sum_j \frac{1}{d_{ij}} \right)^{-1}$$

- **Common Average Referencing (CAR)**

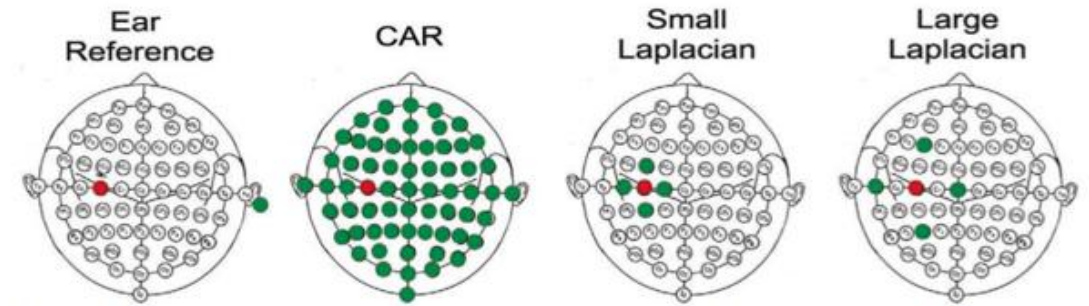
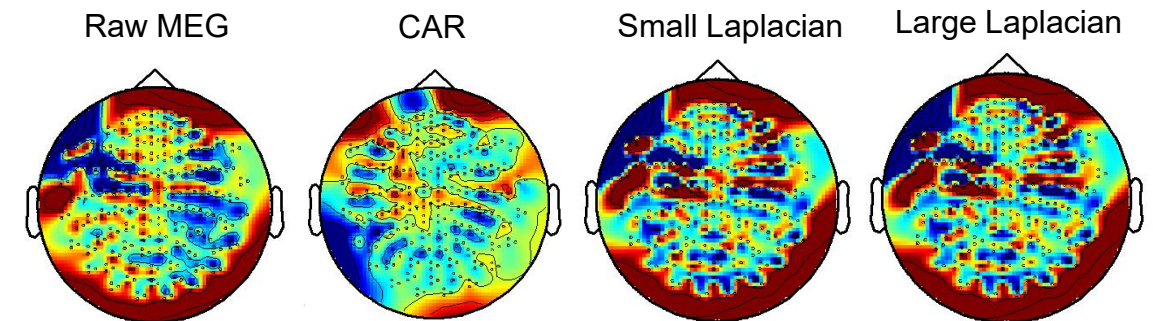


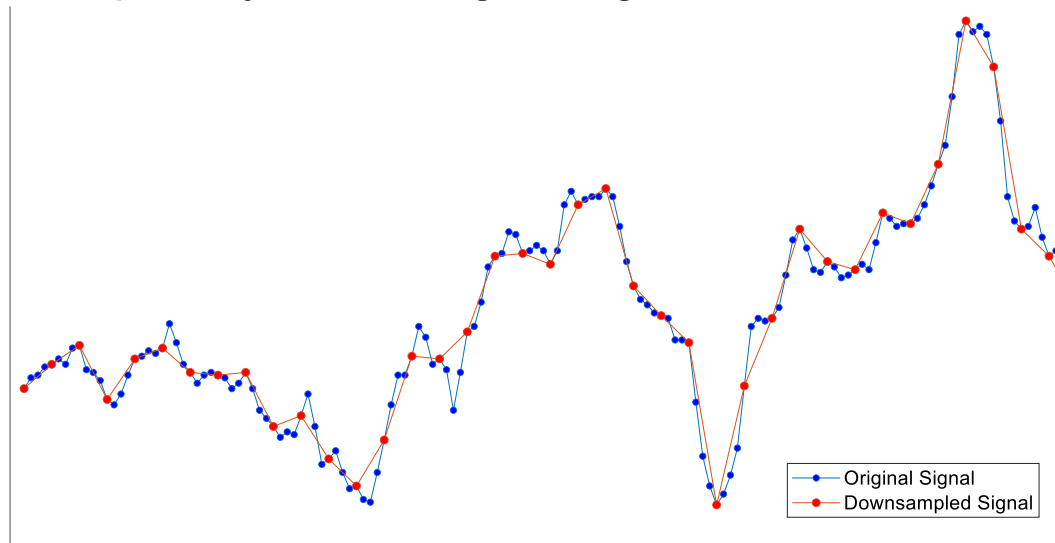
Figure 3. For a clinical EEG array, a mean or weighted mean of green electrodes would be subtracted from the red electrode for each spatial filter listed [7].

McFarland, 1997

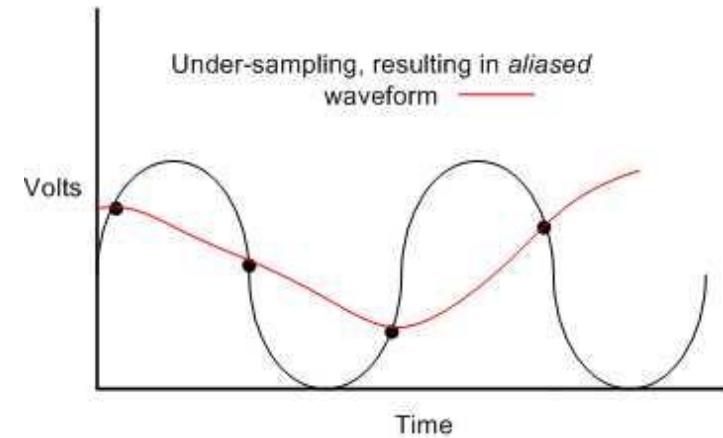
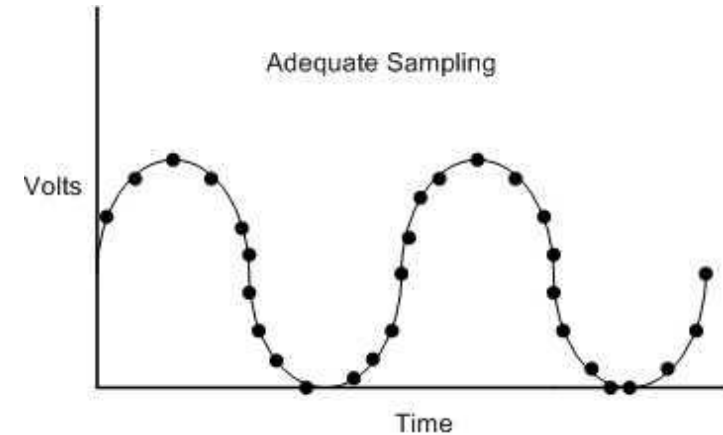


Downsampling

Nyquist Theory – minimum digital sampling frequency must be $>$ twice the maximum frequency in analogue signal

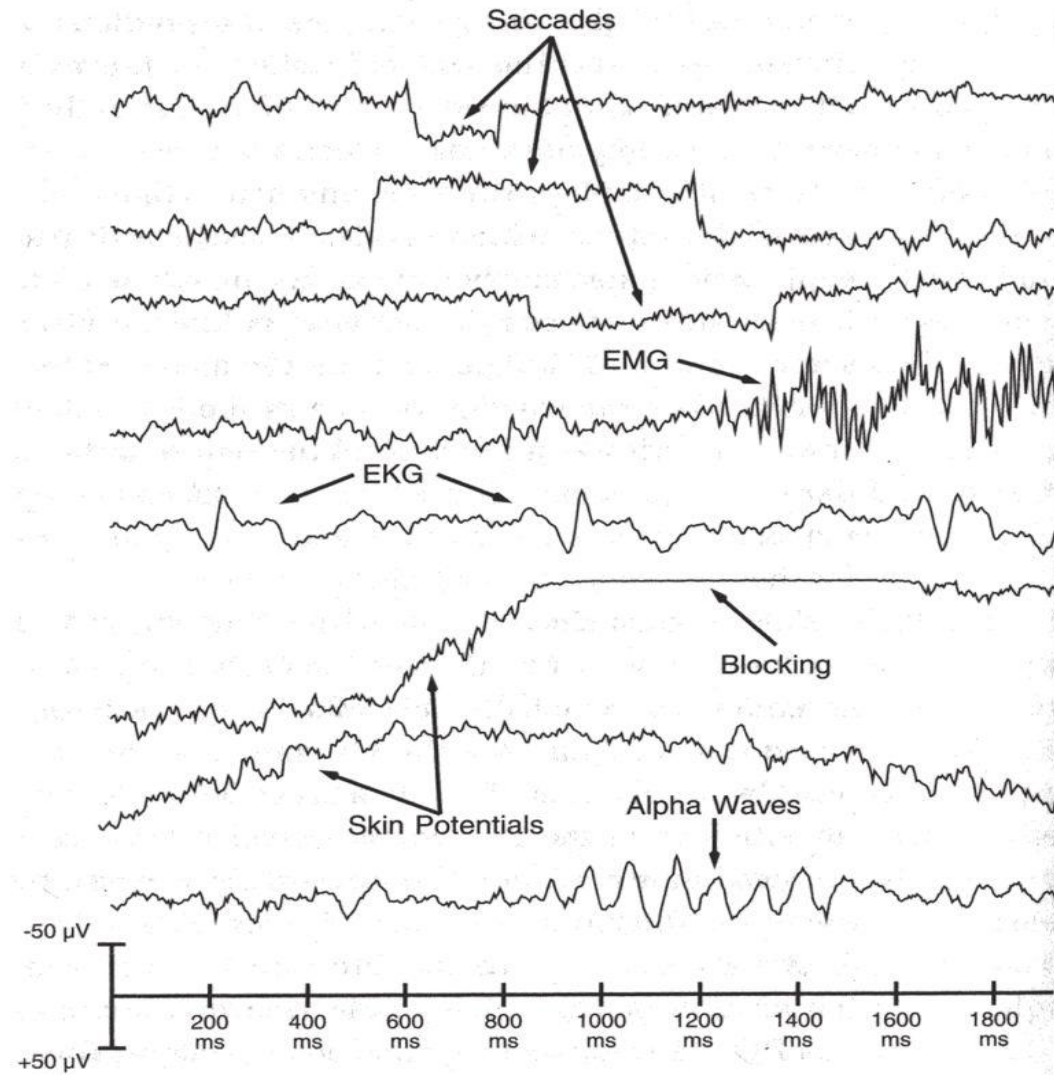


Example of downsampling



Artefact Removal

- Blinks
- Eye-movements
- Muscle activity
- EKG
- Skin potentials
- Alpha waves



Artefact Removal

- **Blinking**
 - Avoid contact lenses
 - Build 'blink breaks' into your paradigm
 - If subject is blinking too much – tell them
- **EMG**
 - Ask subjects to relax, shift position, open mouth slightly
- **Alpha waves**
 - Ask subject to get a decent night's sleep beforehand
 - Have more runs of shorter length – talk to subject in between

Artefact Removal

EOG/Blinks

- most common contaminants of the EEG signal.

Linear Regression

- The main assumption in this approach is that each EEG channel can be expressed as the sum of noise-free EEG signal and a fraction of the source artifact available through EOG electrodes.
- Let S be the recorded EEG signal which can be expressed as the sum of noise-free EEG signal E and EOG or eye blink signal B multiplied by a weight matrix W .

$$S = WB + E$$

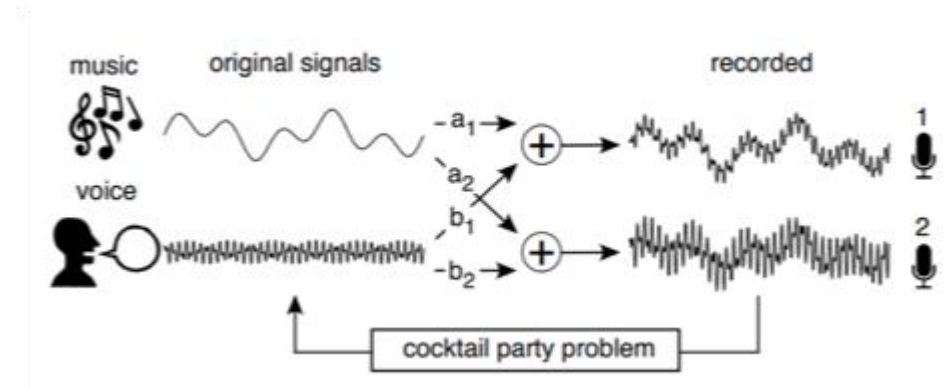
W describes the contribution of the EOG artifact in each EEG channel

Artefact Removal

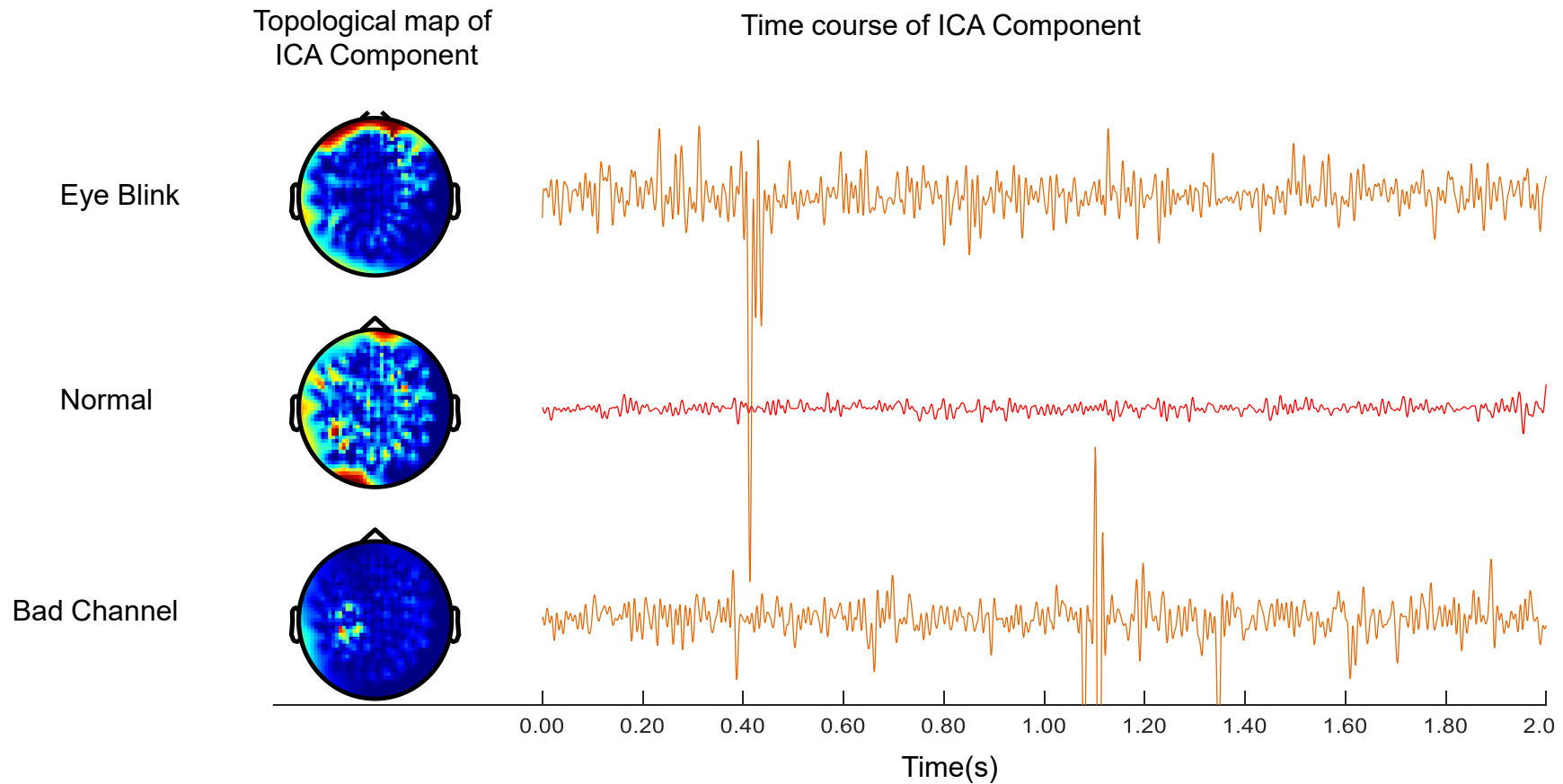
EOG/Blinks

Independent Component Analysis

- Independent component analysis (ICA) is a blind source separation (BSS) technique that is widely used in an array of signal processing applications.
- Once the components have been identified, to remove the EOG artifacts, one can visually determine which independent component corresponds to eye-blinks or movements based on the following criteria.



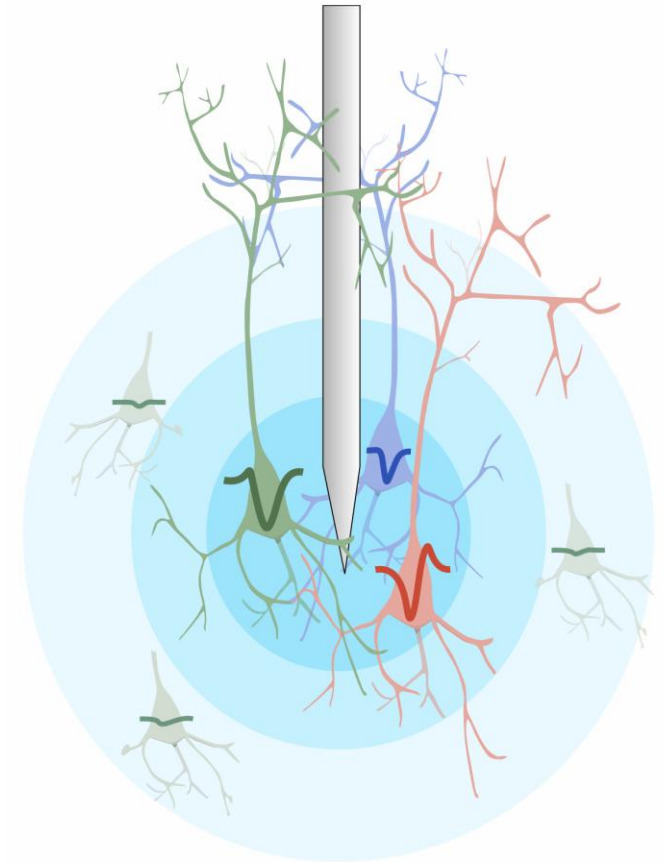
Independent Component Analysis



Common Signal Processing Approaches by Modality

Spike Sorting

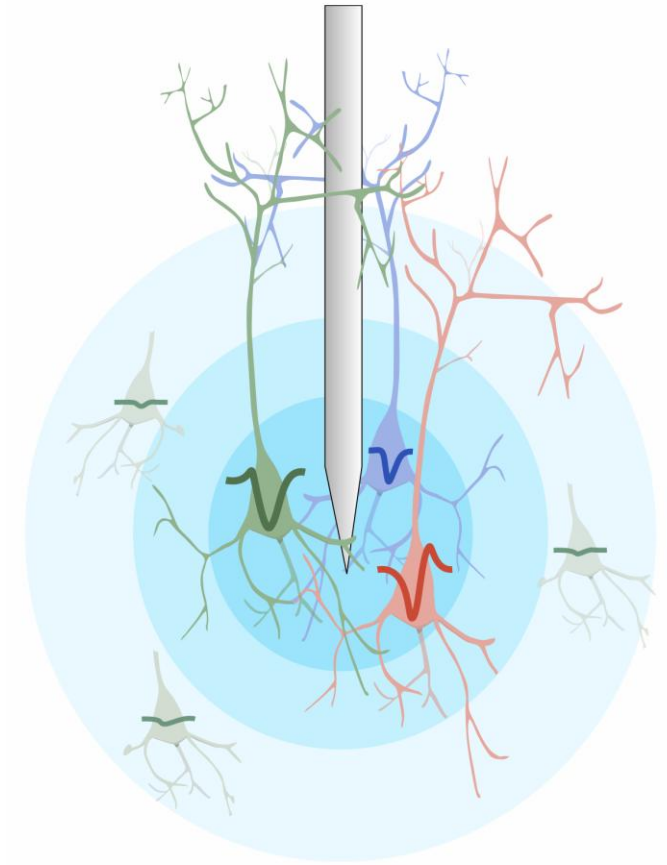
- Spike sorting is the process of **separating and identifying action potentials (“spikes”)** from **different neurons** recorded by the same electrode (or electrode array).
- Each spike has a slightly different **waveform shape** depending on:
 - The neuron’s distance from the electrode tip
 - Neuron size and orientation
 - Electrode properties
- Because of this, you need to sort out *which spike came from which neuron*.



https://en.wikipedia.org/wiki/Spike_sorting

Spike Sorting

- **Why is it important?**
 - Allows you to attribute spikes to **single neurons** rather than “multi-unit” activity.
 - Essential for studying neural coding, plasticity, and brain–machine interfaces.
 - Without sorting, you’d lose information about individual neurons’ firing patterns.



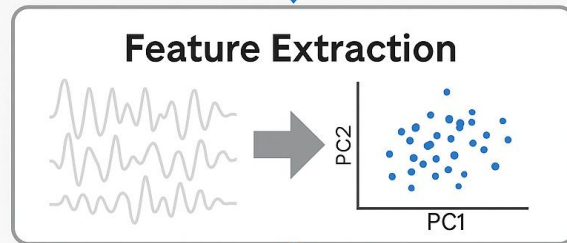
https://en.wikipedia.org/wiki/Spike_sorting

Spike Sorting

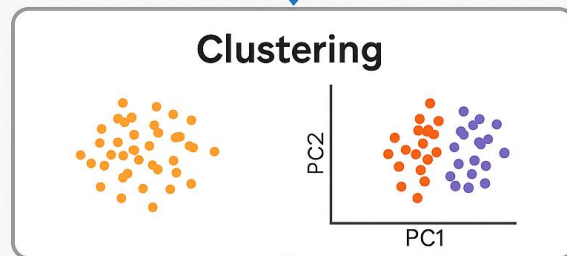
- Steps



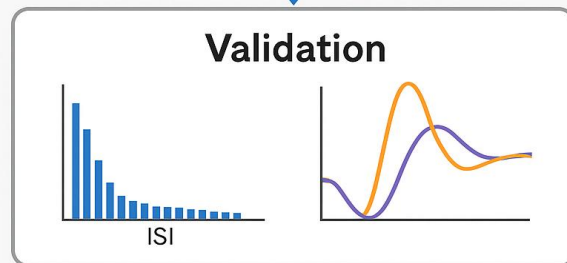
Generally, follow a threshold rule



Principal Component Analysis



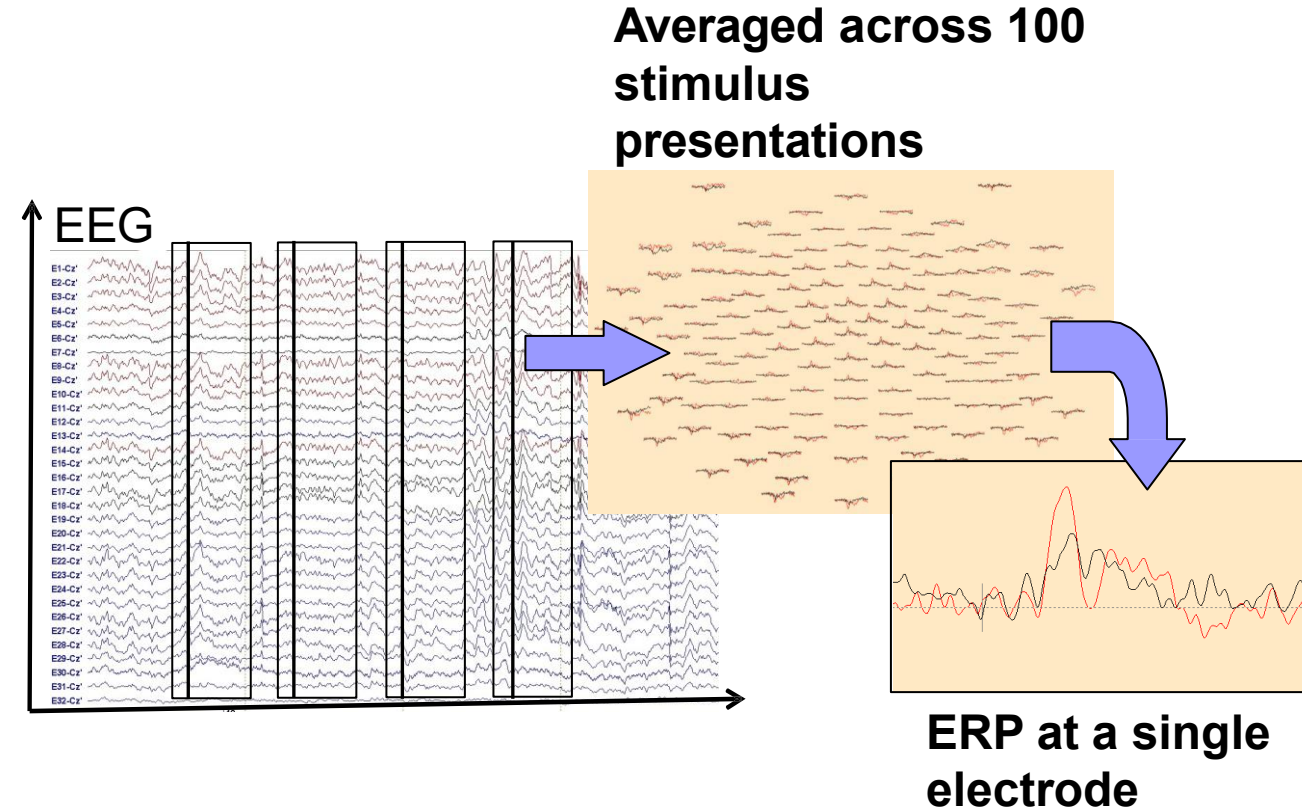
Gaussian Mixture Model, K-Means, Generative Model



Event Related Potential

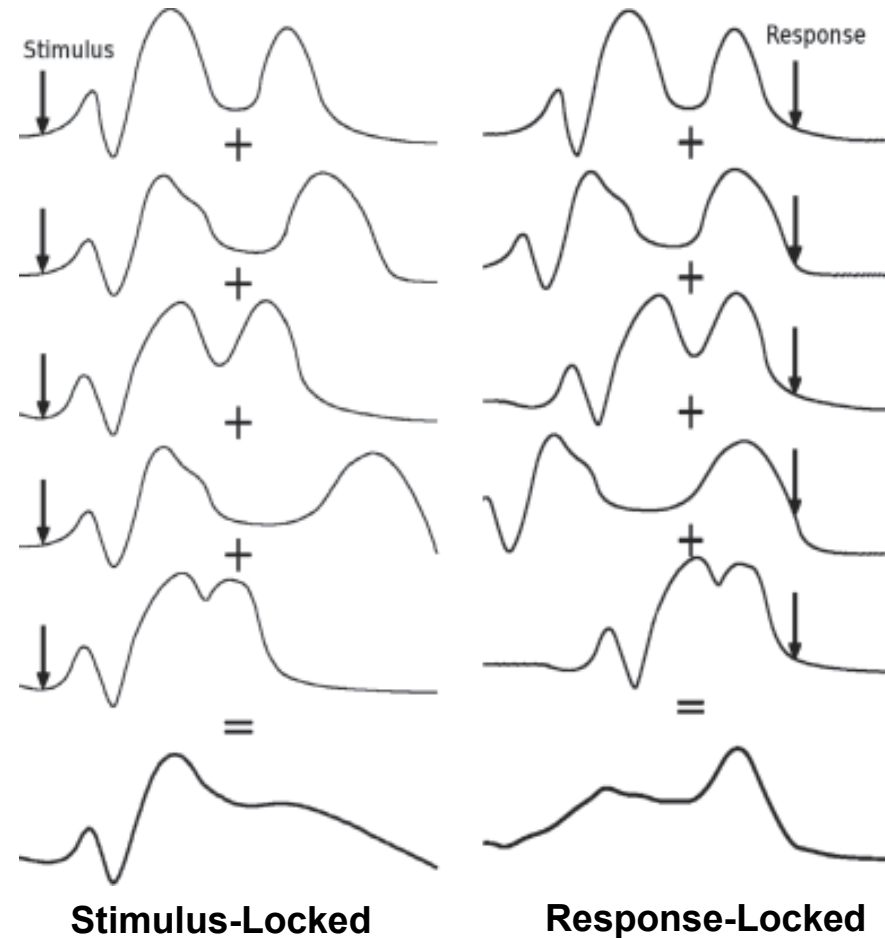
Mostly for EEG/LFP signals

- The averaging technique is used to study the electrical activity time-locked to an event.
- Needs a considerable number of trials
- Comprises a mixture of different brain rhythms, depending on the filters applied.
- Only about 20% of the evoked activity is shown
 - Other approaches to study electrical brain activity: Time-frequency analysis



Event Related Potential

Stimulus-Locked and Response-Locked

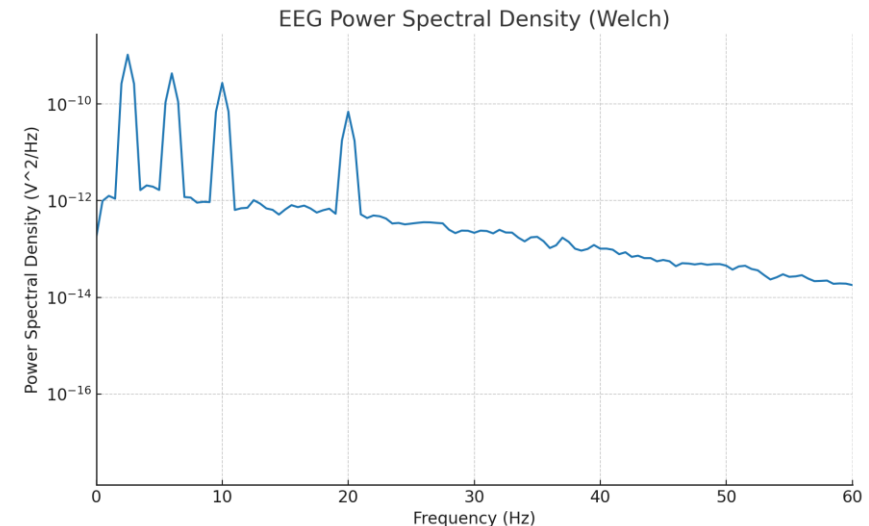


Power Spectral Density

Frequency domain analysis

In practice, PSD is estimated by:

1. Splitting data into overlapping windows
2. Applying a taper or window
3. Computing Fast Fourier Transform of each segment
4. Averaging squared magnitudes



Power Spectral Density

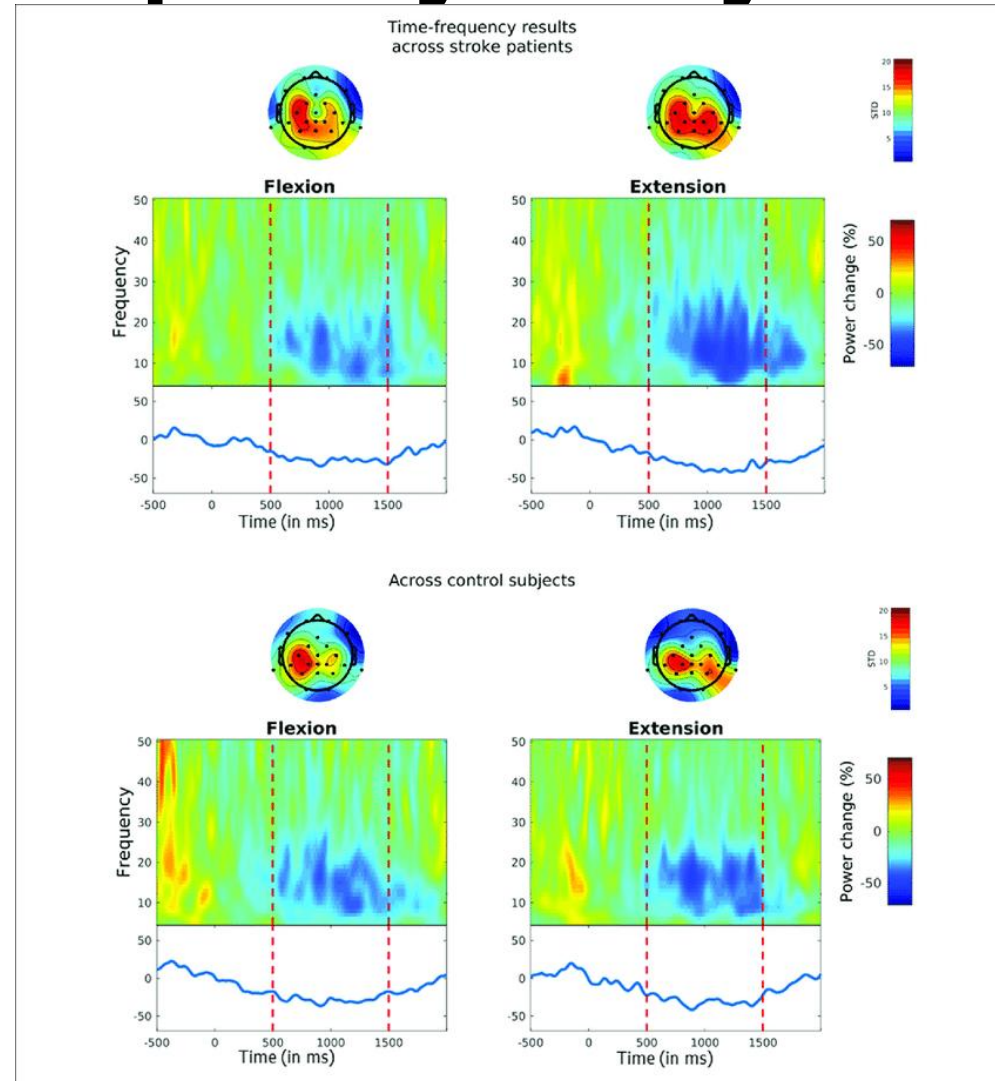
PSD (Power Spectral Density) shows how EEG power is distributed across frequencies.

Why is it useful?

- Reveals **brain rhythms** (delta, theta, alpha, beta, gamma).
- Tracks **neural states** (e.g., sleep, attention, memory load).
- Allows **comparisons across conditions** (e.g., alpha power eyes-open vs eyes-closed).
- Detects **pathological patterns** (epilepsy spikes, Parkinson's beta).
- Provides **features for BCI/ML** (band power, peak frequency).

Interpreting Time-Frequency Analysis

Event-related desynchronization (ERD) during MI. Time-frequency (TF) plots show the percentage change in power from baseline (i.e. from -0.5 s to 0 s) for MI flexion trials (left panels) and MI extension trials (right panels). MI started at time point zero and was performed for 1.5 s. Vertical lines indicate the chosen time interval for the statistical analysis (i.e. from 0.5 s to 1.5 s). The solid blue line on the bottom reflects MI-related power changes within the 10 – 25 Hz SMR frequency range.



Common Brain Signal Analysis

- **Time Domain** – Event-Related Potentials, Amplitude & Latency Measurement, Single Trial Analysis, Cross-Correlation & Coherence
- **Frequency Domain** – Power Spectral Density, Fast Fourier Transform, Spectral Coherence , Cross-Spectral Density , Event related Spectral Perturbation, Band Power
- **Time-Frequency Domain** – Short-Time Fourier Transform, Inter-trial Coherence, Cross-Frequency Coupling
- **Brain and Functional Connectivity**

EEG & LFP

Some interesting papers to read

- Sun, C. & Mou, C., 2023. Survey on the research direction of EEG-based signal processing. *Frontiers in Neuroscience*, 17, Article 1203059. doi:10.3389/fnins.2023.1203059
- Zhang, H., Zhou, QQ., Chen, H. *et al.* The applied principles of EEG analysis methods in neuroscience and clinical neurology. *Military Med Res* **10**, 67 (2023). <https://doi.org/10.1186/s40779-023-00502-7>
- A. Jackson and T. M. Hall, "Decoding Local Field Potentials for Neural Interfaces," in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 10, pp. 1705-1714, Oct. 2017, doi: 10.1109/TNSRE.2016.2612001.
- <https://buzsakilab.com/content/PDFs/BuzsakiKoch2012.pdf>

Useful Tools

Toolkit	Domain	Platform	Strengths
SpikeInterface	Spiking, LFP analysis	Python	Unified pipeline, sorting, benchmarking, reproducibility
Chronux	Spike, LFP, EEG/MEG	MATLAB	Spectral methods, GUI, statistical rigor
MNE-Python	EEG, MEG, iEEG/ECOG, fNIRS	Python	End-to-end analysis, source localization, tutorials
EEGLAB	EEG/MEG	MATLAB	ICA, plugins, visualization, scripting + GUI
FieldTrip	EEG, MEG, LFP, Spike	MATLAB	Advanced analysis, connectivity, scripting
Brainstorm	EEG/MEG, LFP	MATLAB / GUI	GUI-friendly, reproducible, no MATLAB required
NBT	EEG/MEG biomarkers	MATLAB	Biomarker focus, database, EEGLAB integration

Reading

Why?

- No single method captures both WHERE and WHEN brain activity happens
- Combining modalities provides a more complete picture of brain function

Key Combinations

- **EEG + fMRI** → *ms timing + mm spatial maps*
- **MEG + fMRI** → *better source localization + spatial mapping*
- **Invasive (LFP/Unit) + fMRI/EEG** → *link neurons ↔ networks*

Applications

Cognitive neuroscience (fast rhythms + networks),
Clinical (epilepsy, Parkinson's), Brain–Computer
Interfaces, Cross-validation of neural signals

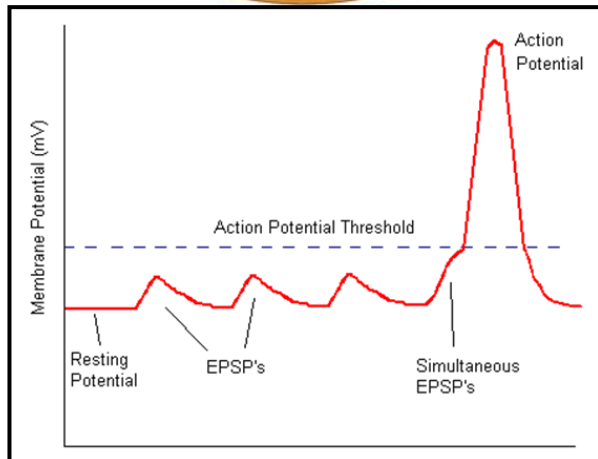
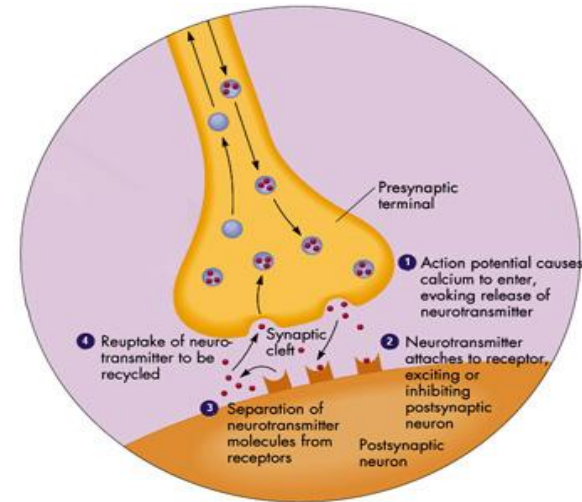
Multi-modal = “Where + When” → a complete view of brain function.

Comparison

Method	What it Measures	Spatial Resolution	Temporal Resolution	Invasiveness	Common Uses
Extracellular Recording	Action potentials from individual neurons	Microns	<1 ms	Invasive	Animal studies, BMIs
LFP	Local synaptic activity	~0.1–1 mm	1–10 ms	Invasive	Oscillations, circuits
EEG	Scalp electrical potentials	cm	ms	Non-invasive	Epilepsy, sleep, BCI
MEG	Magnetic fields from neural currents	mm–cm	ms	Non-invasive	Source localization, cognition
fMRI	BOLD (blood oxygen) response	mm	s	Non-invasive	Functional mapping, networks

Electroencephalography

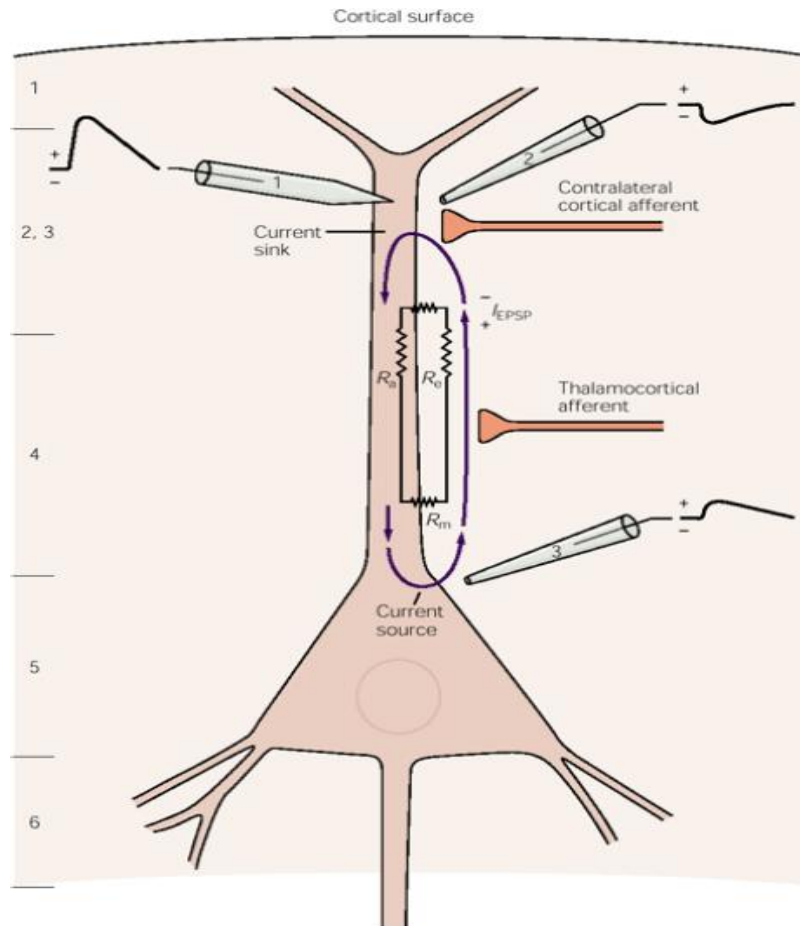
Neural Basis



- 1) When an Action Potential reaches the axon terminal the neuron releases a neurotransmitter.
- 2) The neurotransmitter binds to the receptor.
- 3) The postsynaptic neuron gets depolarized (Na^+ inward currents – excitatory - EPSP) or hyperpolarized (Cl^- inward currents – inhibitory -IPSP).
- 4) EPSP and IPSP summate temporally and spatially.
- 5) If the postsynaptic neuron reaches a given depolarization threshold, an action potential is generated.

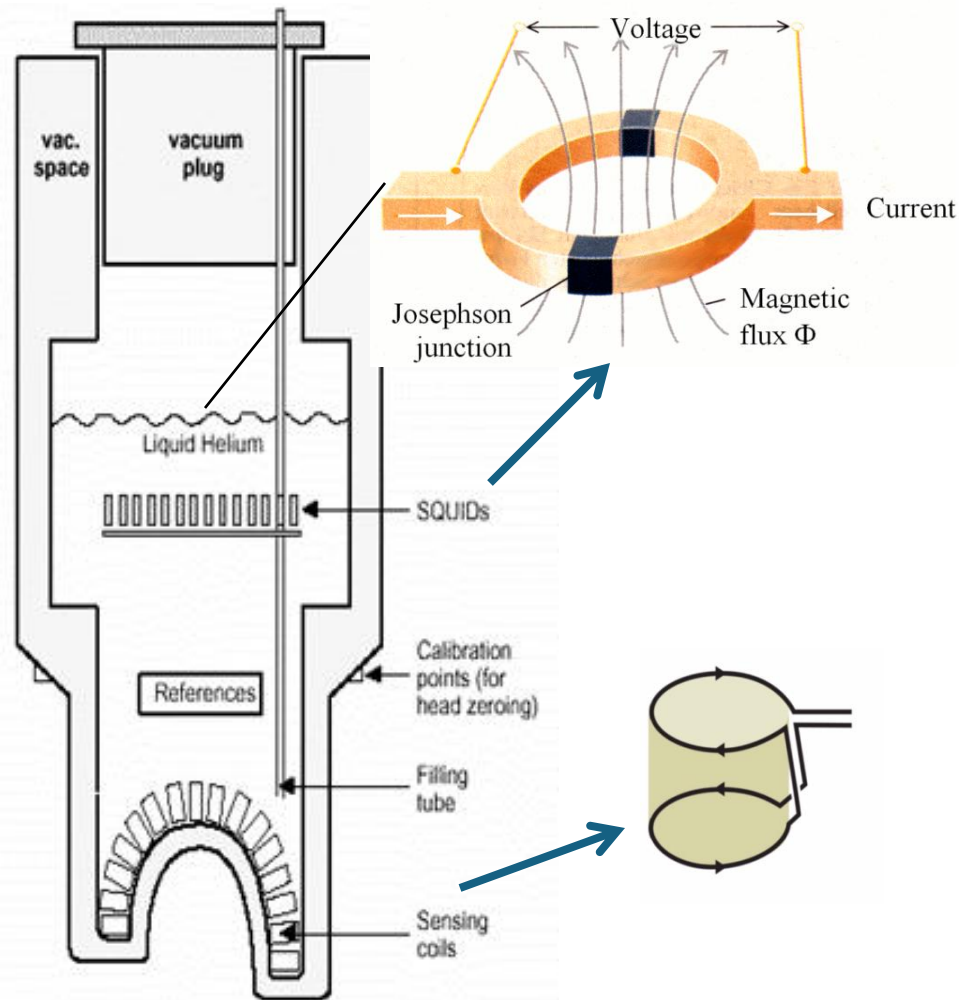
Electroencephalography

Neural Basis



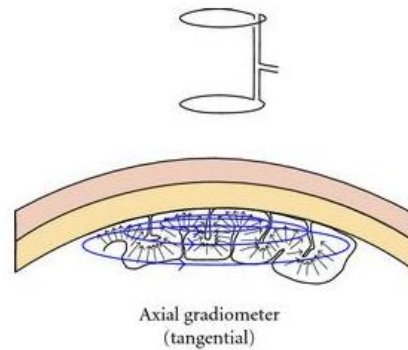
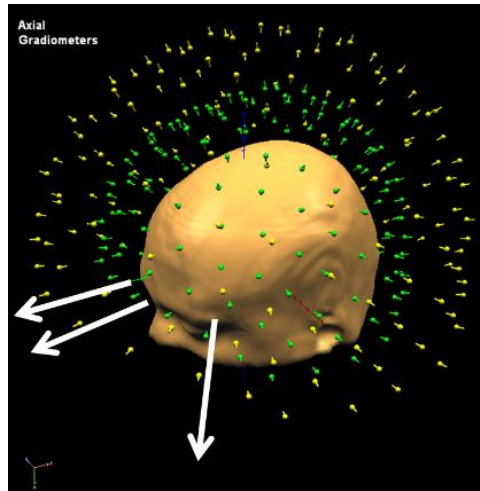
- When an EPSP is generated in the dendrites of a neuron an extracellular electrode detects a negative voltage difference, resulting from Na^+ currents flowing inside the neuron's cytoplasm.
- The current completes a loop further away the excitatory input (Na^+ flows outside the cell), being recorded as a positive voltage difference by an extracellular electrode.
- This process can last hundreds of milliseconds.
- **Thus, a small dipole is generated!!**

Magnetoencephalography



- **SQUID** - Superconducting **QU**antum Interference **D**evice, immersed in super-cool liquid helium
- Sensitive to field changes in order of femto-Tesla (10^{-15})
- Superconductive ring with two Josephson junctions
- Flux transformers (coils)
 - Magnetometers
 - Gradiometers (planar/axial)

Magnetoencephalography



Axial Gradiometer MEG sensors...

- ...are aligned orthogonally to the scalp
- ...record gradient of magnetic field along the radial direction

Planar Gradiometer MEG sensors...

- ...two detector coils on the same plane
- ...have sensitivity distribution similar to bipolar EEG setup

